

# GAURAV SINGH

Portfolio | Leetcode | +91-7352699553 | b24ee1020@iitj.ac.in | LinkedIn | GitHub

## EDUCATION

Indian Institute of Technology Jodhpur

B.Tech in Electrical Engineering | CGPA: 7.66/10.00

2024 – 2028

Jodhpur, Rajasthan

New Horizon Public School

Class XII | 95.4%

2023

Panvel, Maharashtra

## TECHNICAL SKILLS

**Languages:** C++, C, JavaScript, Python(basics)

**Core CS:** DSA, Operating Systems (OS), Object-Oriented Programming (OOP), Database Management Systems (DBMS)

**Backend:** Go(Gin), Node.js, Express.js, REST APIs, WebSockets

**Frontend:** React

**Databases:** PostgreSQL, Redis

**Systems:** TCP/IP, Socket Programming, Multithreading, Concurrency, Synchronization

**Tools:** Git, Linux, Docker, Postman

## PROJECTS

**Mini Redis: Redis-Inspired In-Memory Database** | C++17, TCP Sockets, Multithreading, STL, File I/O

[GitHub](#)

- Architected a Redis-inspired **high-performance in-memory key-value store** in C++17, serving concurrent TCP clients via an **8-thread producer-consumer worker pool** with mutex and condition variable-based thread synchronization.
- Introduced data durability through **snapshot persistence and Write-Ahead Logging (WAL)**, achieving **crash recovery of 100K keys in 170ms** via log replay and checkpoint-based restoration.
- Integrated **active and lazy TTL-based key expiration** with **O(1) LRU cache eviction** using hash maps and doubly linked-list indexing, sustaining **50K+ insertions** under enforced memory capacity constraints.
- Created a custom **RESP protocol parser** supporting Redis-style commands (SET, GET, DEL, EXPIRE, TTL, COUNT), achieving **95K ops/sec** under concurrent load and **35K GET ops/sec** in single-client benchmarks.
- Enabled runtime-configurable server parameters via external config files; performed **benchmark-driven performance profiling** across throughput, latency and crash-recovery scenarios.

**Real-Time Collaborative Code Platform** | Go (Gin), React, PostgreSQL, Docker, WebSockets, Monaco Editor

[GitHub](#)

- Developed a **large-scale real-time collaborative coding platform** in Go enabling concurrent multi-user sessions with live code synchronization, presence tracking and shared execution output over **WebSockets**.
- Established a **scalable room-based session model** with **PostgreSQL-backed persistence** and **JWT-authenticated REST APIs**, enabling secure workspace creation, rejoining and state continuity across sessions.
- Provisioned **sandboxed multi-language code execution** (Python, JavaScript, C++) using **Docker containers** with enforced CPU, memory, network and time-limit constraints for **secure isolated runtime environments**.
- Integrated Monaco Editor with **low-latency bidirectional WebSocket communication**, delivering an IDE-like collaborative experience with real-time participant tracking and synchronized execution results.

**AsyncTasker: Fault-Tolerant Distributed Job Queue** | Node.js, Express.js, Redis, PostgreSQL

[GitHub](#)

- Developed a **distributed, fault-tolerant background job processing system** using **Redis-backed priority queues** and **PostgreSQL persistence**, ensuring **at-least-once task execution** under high concurrency, sustaining **1,093 jobs/sec** with a 10-worker configuration.
- Designed a **concurrent worker pool** with configurable parallelism, active-job tracking, timeout enforcement and graceful job lifecycle management across failure scenarios, delivering **sub-10 ms P99 API latency**.
- Incorporated **priority-aware delayed scheduling** and **exponential-backoff retry logic** using **Redis Sorted Sets**, enabling resilient recovery from transient failures and reducing job loss under partial outages.
- Established automated **crash-recovery workflows** that reconstruct in-memory queue state from persistent storage and reclaim **abandoned in-flight jobs** after unexpected worker or service restarts, restoring **10K jobs at over 2,176 jobs/sec**.

## ACHIEVEMENTS & PARTICIPATION

- Solved **500+** DSA problems across Coding Platforms - Leetcode, GfG, Codeforces
- Leetcode Contest Rating : **1600+**
- Secured **1st Rank** in School in CBSE Class XII Board Examination

## CAMPUS INVOLVEMENT & LEADERSHIP

- **Core Member, Programming Club**, Co-organized competitive programming contests and workshops.
- **Student Guide, Student Well-being Committee**, Mentored 14 Freshers both academically and non-academically
- **Support Team, CDC**, Helped Placement Co-coordinators in organizing placement processes